

Holistic User Modeling in Smart Environments

The Case of the As-A-Pal Project

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Abstract As-A-Pal is a holistic approach for seamlessly combining generic medical and health knowledge with specific information about a user's needs, preferences, environment, activity patterns, etc. This is accomplished by means of i) different sources of information (e.g., applications utilizing sensor networks, medical knowledge repositories and assessment applications); ii) a common semantic model, i.e., vocabulary as base for holistic ubiquitous user modeling, which is shared between applications, and iii) the technology developed for fusing information and for reasoning about the status of the user, about decisions to make and about what actions to take, partly by software agents in their managing of the user model. The contributions are described and exemplified from the perspectives of archetype user profiles (personas) with cognitive and/or psychiatric conditions such as depression, anxiety or social withdrawal.

A pilot evaluation study was conducted focussing the motivational structures and how the user model was modified leading to tailored assessments. The study provided results that are fed into the redesign and development of the multi-agent system, which manages the user model and tailors the support provided the user.

Keywords Knowledge-based systems · User modeling · Ubiquitous computing · Pervasive computing · Human-computer interaction · Ambient assisted living · Ontology · Possibilistic argumentation · Activity theory · Multi-agent systems · Mental health · Behavior change systems · Participatory action research

1 Introduction

We address some of the current challenges for ubiquitous user modeling, with a particular focus on *knowledge-based support* for users' activities in smart envi-

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ronments (*i.e.*, human interaction with intelligent services). Fusing information from heterogeneous sources offers challenges like *privacy*, *trust*, *conflicts* and *uncertainty* that should be dealt with at a semantic level (*e.g.*, [9, 21, 8]). Moreover, the notion of the *context* of the user, which is complex and changing, was identified as the main challenge for ubiquitous user modeling by Kuflik and coworkers [28]. Smart services, that are personalized to a human's needs and desires, require higher-level reasoning with information and knowledge also beyond the scope of an individual. The surrounding physical (including data capture devices for ubiquitous computing), social and knowledge context, which may enable, as well as constrain human activity should be included in ubiquitous user modeling.

In addition to information about the environment, the ongoing activity and the individual's preferences and needs, there is a need for integrating generic medical and health knowledge for an ambient system to be responsive to the individual's needs. For instance, one reason for including health-related knowledge in a holistic approach to ubiquitous user modeling is the decreasing well-being among young individuals between the age of 18-24 years old, in particular young women. In the year 2011 10 % of young women and 7 % of young men was subjected to pharmacological treatment and/or hospitalization due to a psychiatric condition [52]. The decrease of well-being is manifested in high levels of perceived stress and work load, and a dramatic increase in pharmacological treatment for attention deficit disorders, depression and anxiety [50]. The behavior a young person adopt for managing these conditions can be increase of activities to take control over one's situation, by high performance in school, physical and social activities, sometimes to a level where the behavior becomes unhealthy. Other coping strategies can be self-destructive behavior such as self-injuring, eating disorders, drug abuse or social withdrawal (*e.g.*, [49]), for the purpose to decrease their anxiety.

Another reason for integrating health-related knowledge is the involuntarily isolation at the other end of the life span, which older adults commonly face after retirement from working life, and the situation caused by a range of conditions emerging by age, both medical and psychiatric, which are tightly intertwined in an older individual. In a study of the older adults in Sweden aged 65-75 who was treated for a psychiatric disease in 2011, it was shown that in about 50 % of those who was diagnosed having depression or addiction to alcohol or other drugs, this condition started after the age of 65 when most people retires in Sweden. The same onset pattern was shown also in about 80 % of those who suffered from an anxiety disease [51].

We recognize that these conditions are common, both triggered and inhibited by circumstances in the social and cultural context and may affect anyone at some point in their lives (*e.g.*, [1]). Moreover, they are highly intertwined with activity performance and motivation, thus essential to take into consideration in ubiquitous user modeling and in reasoning about appropriate and tailored support to individuals.

Against this background, we put forward the As-A-Pal architecture, which is an architecture aiming at providing Agent-Supported Assessment for Per-

sonalised ambient Assisted Living. As-A-Pal is a holistic approach for seamlessly integrating generic medical and health domain knowledge with specific knowledge about an individual's environment, activities, needs and preferences obtained from different sources, for the purpose to tailor the support for accomplishing chosen activities (work, leisure, social activities, *etc.*). The motives are to increase ability, autonomy, knowledge, wellbeing, social inclusion and health in an individual.

The focus of this article is on the holistic approach to ubiquitous user modeling applied in As-A-Pal, which follows the therapeutic assessment and intervention procedure. The procedure typically consists of the following. An initial assessment is conducted, a holistic understanding of the client is established (*i.e.*, a *client*, or *user model*), a plan of interventions is made, implemented and followed up with adjustments of both the client model and interventions according to detected changes in needs, ability, health and preferences.

If active and healthy living and aging is aimed to be promoted, the activities perceived as purposeful and desired by the individual need to be identified and supported [25,42]. Moreover, this adaptivity should be provided on the basis of conditions defined and regulated by the individual [9]. Therefore, we address the following questions:

- How does the user model capture motivational structures?
- How is intervention established based on the initial assessments?
- How is the information obtained from the ambient environment contributing to the user model, initial assessments, the planning of interventions and followup assessments?

In order to give answers to these questions, the As-A-Pal architecture is introduced. This architecture combines an ambient assisted living environment with continuous computer-supported and computer-guided assessments lead by health care professionals. Moreover, this architecture combines different methods for personalization, such as methods for *adaptation* (by the user, family members or health care professionals), and *adaptivity* (semi-autonomous interpretation of the user's activity patterns and decision-making about intervention). These two types of channels of information about the user, which both may generate a revision of the user model, are fused into a *common semantic model* based on an *actor and activity ontology*. This semantic model can be understood by the human actor and is computable by the system agents. Services requested by the individual are provided based on this architecture.

As-A-Pal is a development of the framework presented in [34], and integrates different mechanisms enabling a personalized and dynamic user model, which are introduced in Section 3. These include a *core ontology* as base for holistic user models and for the common vocabulary used for fusing information from the different sources of information (Section 4). The agent-based technology for reasoning about the status of the user, about decisions to make and about what actions to take, *i.e.*, management of the user model is presented in Section 5.

As-A-Pal is applied in a case study for the purpose to investigate how the motivational structures are captured by the user model, giving the user the opportunity to define and prioritize their goals for improving their situation. The evaluation study is presented in Section 6.

The approach is discussed in the context of related work in Section 7 and in relation to challenges when using computer-based support for promoting and supporting a persistent behavior change in Section 8, before the contributions are summarized in Section 9.

The following section gives an overview of the methods applied, and theories underpinning our work.

2 Methods and Theoretical Framework for Holistic Models of Users

The main goal and starting point for our work is to disseminate evidence-based medical and health knowledge and best practice knowledge to individuals in a context where it is useful, and tailored to individuals. Examples of integrated knowledge are guidelines for assessing cognitive impairments and disabilities performing daily activities, and protocols for assessing risk for falling in a home environment. The reason for using established domain knowledge is for optimizing quality in the health services provided the user. Consequently, a knowledge-based user modeling approach is required, which is combined with methods for identifying specific individual features [28]. The domain knowledge was modeled and integrated by domain professionals in an end-user development process using the ACKTUS platform [33].

Ontologies were developed for representing the knowledge about the domain, the user and the context [35]. For the purpose to capture information about activities obtained from ambient environments, the ontologies were further developed as part of this work. The content of the ontologies is used for reasoning, partly using theories and methods based on formal argumentation frameworks. This for the purpose to capture uncertainty, ambiguities and inconsistencies in the fusion of heterogeneous sources of data [27].

Activity Theory [25] and the Model of Human Occupation [26] are two systemic models for describing humans performing purposeful activities, and the factors influencing activity execution. They have been applied in the design of the ontologies, and in the design of the reasoning engines.

A participatory action research methodology is being applied in the development of the knowledge engineering platform ACKTUS and the different knowledge-based applications and services integrated in the As-A-Pal framework.

In the development of different components of As-A-Pal we focused our requirements analysis on *personas* and a case scenario (e.g., [12,11]). A persona is an archetype of a potential user, which functions as a knowledge artifact in the development process. The main benefit of applying this approach is that it allows for sharing a common goal scenario among all professionals participating in the development and across professional domains. Our scenario is a

description of how the anticipated use situations may proceed when technology is developed to support the described situations [11].

The main scenario is designed to capture a holistic perspective on assessment in a use situation (i.e. from a health care professional's perspective), which makes our scenario different from e.g., [2]. The intervention part of the scenario gives also detailed descriptions of the desired activities to be performed from the perspective of the persona. We used the personas and scenario in the development of the ACKTUS actor and activity ontology, domain knowledge and applications, for specifying the functional requirements of the multi-agent systems, for the smart kitchen environment and as baseline for the evaluation study design. We describe the personas in more detail in the following subsection.

2.1 Personas and Use Scenarios

The motivational structures and the reasons for limitations in performing activities vary between individuals of different age, embodying different personalities, coping strategies, physical, cognitive, psychological, social and physical abilities. Therefore, we identify four different core personas representing recognized phenomenon present in our current society. The four personas are described in Table 1.

The persona and scenario of the main older adult we call Rut, was created based on eight older adults participating in a pilot study on personalized support systems, which resulted in the initial prototype of the I-Help application, and the outline for the As-A-Pal architecture [32,34]. For the purpose to address a broader range of potential users and complex situations, we added as part of this work an additional persona of an older adult, and two personas and their scenarios, which represent young adults who's health would benefit from a change of behavior. The two young personas were created based on interviews with four occupational therapists specialized on young people with neuropsychiatric disorders. Two of these have particular knowledge on assistive technology for compensating for limitations in cognitive functions.

Rut is previously presented in [34] and is suffering from a pain condition, mild depression, anxiety partly due to her fear of falling down, and mild cognitive impairments. By contrast, the persona Greta is comparatively healthy, and the things they have in common are their age, their living situation, their social network, and a physical impairment since Rut has difficulties walking due to her pain condition, and Greta has difficulties walking and using her left arm and hand due to the remaining impairments from a stroke a few years earlier.

We are contrasting the two older adult personas with the scenarios of two young female personas with some common features such as similar age, background and living situation. One persona who we call Eva has had some difficulties coping with her teenage period, causing mild depression and anxiety. She is diagnosed suffering from an attention deficit disorder in addition to anx-

Table 1 Personas and their characteristics.

Persona	Strengths and Limitations	Interests	Goals
<i>Rut</i> 84, living alone.	Physical impairment: pain conditions, walks with aid. Has worries about falling, mild depression, mild cognitive impairment. Is becoming more and more isolated. Is used to having a socially rich life.	Is interested in politics, what is happening in society, and in socializing with friends and family.	Prevent falling, contact with Greta and her son, keeping up with society.
<i>Greta</i> 80, living alone	Physical impairment: walking with walking aid due to a right hemisphere stroke seven years earlier, pareses in left hand and arm. Is used to having a socially rich life. Is comparably active, trying to maintain her health and social network.	Takes interest in social activities, engaging in organizations, playing instruments and watching TV.	Maintain food, medication routines and other health related issues.
<i>Thea</i> 18, living with parents	is very active, engaging herself in every activity which she finds interesting and fun, in addition to making a good effort getting the school work done.	Takes interest in many things, manifested in going to gym classes, arranging parties with friends, being a volunteer leader for younger teenagers.	Reduce and maintain activity level at a normal and healthy level, including physical exercise.
<i>Eva</i> 18, living with parents	is isolating herself at home and is having difficulties initiating and organizing school work. Diagnosed having a limiting level of anxiety, attention deficit disorder and social withdrawal. Is efficient and very focussed at the topics, which interests her. Is less motivated for doing things she finds less interesting.	Takes interest in animals, handicrafts, playing instruments, listening to music, computer games.	Increase social activity level, leave home, maintain school work, medication and food routines.

iety and a tendency of social withdrawal. By contrast, a persona we call Thea is doing very well in school, is very active, engaging herself in every activity which she finds interesting and fun. This she does to the extent that she is on the border to exhaustion. Both are considered having an "unhealthy" pattern of behavior, and the aim is to provide both support for changing their pattern of behavior towards a healthier way of approaching their daily activities.

These four personas are for different reasons limited in their daily activities, to the extent that they can be subjected to different kind of diagnoses for explaining their limitations in performance from a societal perspective. However, from a therapeutic perspective, the "pathology" is not in focus, instead strengths, abilities and their environments, which can be used for compensating for and challenging limited ability. The personas will be used for exemplifying the holistic approach for user modeling in the following sections.

3 Introducing ACKTUS and As-A-Pal

ACKTUS (Activity-Centered Knowledge and interaction modeling Tailored to USers) is an evolving *semantic web application* that is designed to allow domain experts who are typically not familiar with knowledge engineering to author and model the knowledge content of, and design the interaction with, knowledge-based applications [36]. The knowledge-based support applications described and used in our case study were developed by domain professionals using ACKTUS [33].

ACKTUS provides the following key functionalities: editors for developing tools and structures for data capture tasks, and editors for developing knowledge and reasoning content and structures. A toolbox of editors aiding these main tasks is provided containing the following: a procedural statement (s-node) editor, a factual statement (information-node or i-node) editor, a concept editor, a use case editor and a scale editor. The s-node editor allows the domain expert to define patterns of reasoning, which are used by the system agents to create *formal rules* and *arguments*. From observations or data collection tasks in a particular case, new knowledge can be derived when combined with the generic domain knowledge represented as s-nodes, such as *conclusions* about diagnoses, *recommendations* and advices, and suggestions, or the initiation, of *activities*. This structure is further described in Section 4.1.

The As-A-Pal architecture is an expansion of the ACKTUS architecture presented in [36], with embedded modules for activity recognition and interpretation in ambient environments. The architecture contains modules for ambient assisted living, which integrates support for assessments and daily activities (Figure 1). The architecture serves as a platform for knowledge management, ambient assisted living and decision-making, utilized by the ACKTUS applications developed primarily by experts in medical and health domains.

The As-A-Pal architecture is composed by three layers (Figure 1):

Data layer: The aim of this layer is to keep the information which is collected from the interaction layer and inferred from the intelligent layer. The data layer is managed by semantic data repositories. In particular, this layer has been implemented by SESAME repositories¹. Therefore, the information is captured in terms of RDF data (the Resource Description Framework (RDF)²) and OWL ontologies (the Web Ontology Language (OWL)³). The primary semantic model which is kept in this layer is *the ACKTUS semantics model* which is managed as an OWL ontology. This semantic model plays a key role for managing issues of *data interoperability* between the different components of the As-A-Pal architecture (see Section 4.1). Moreover, the ACKTUS semantic model was extended for capturing ambient information which comes from the interaction layer and inferred from the intelligent layer (Section 4.2).

¹ <http://www.openrdf.org/>

² <http://www.w3.org/RDF/>

³ <http://www.w3.org/2001/sw/wiki/OWL>

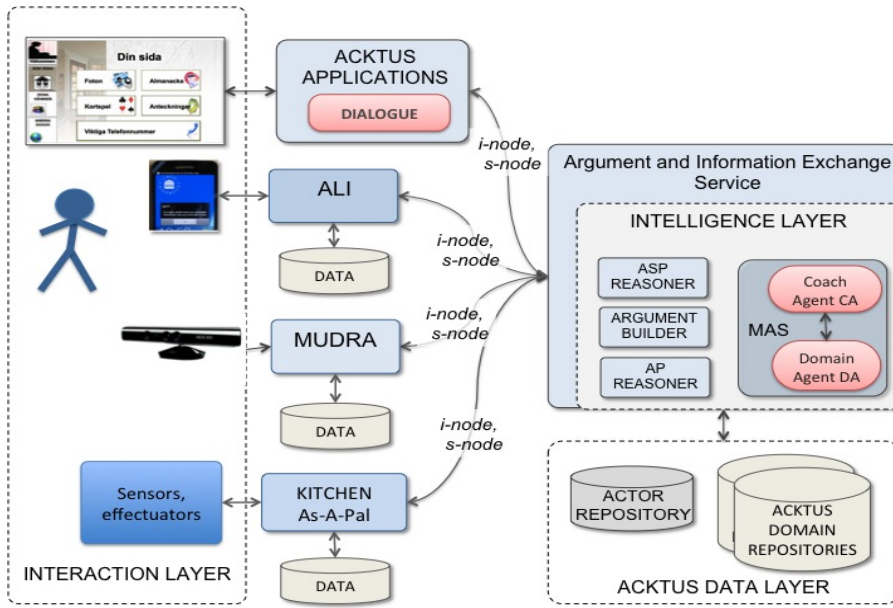


Fig. 1 As-A-Pal architecture

Interaction layer: The main purposes of this layer are to collect information from the user and provide support to the user. In the As-A-Pal architecture, the information about a user is collected from two main types of sources, visible as the *interaction layer* in Figure 1; i) the knowledge-based ACKTUS applications (Section 5.1), and ii) the sensor-based ambient systems embedded in the environment and devices (Section 5.2). Another purpose of this layer is to detect and evaluate the activities, which the client has prioritized using the ACKTUS applications.

Intelligence layer: The main aim of this layer is to support *decision-making*. To this end, two main approaches of intelligent reasoning has been implemented: logic-based reasoning in terms of *Answer Set Programming (ASP)* [4] and *argumentation reasoning* [7]. These two reasoning methods have been regarded as the state of the art in symbolic-based reasoning methods in *Artificial Intelligence* [4,7]. Based ASP and argumentation reasoning, different reasoning engines have been implemented [20,54]. These reasoning engines support reasoning processes such as *human activity recognition* [19] and *medical diagnosis* [54]. These engines have been embedded in FIPA agents⁴. In particular, JADE (Java Agent DEvelopment Framework) has been used for implementing multi-agent systems (MAS).

The key functionalities that are used for holistic user modeling in the As-A-Pal ambient environments are further described in the following sections.

⁴ <http://www.fipa.org/>

4 Using Ontologies for Representing the User in Ubiquitous Environments

The ACKTUS core ontology consists of an *actor and activity (AA) ontology* and a *reasoning and knowledge modeling (RKM) ontology*.

In addition to these, an *Actor repository* is used for storing the information specific for a user in an ontology based on *events*, to which sets of information are associated. The events and information are extracted and put into a semantic context using the ACKTUS core ontologies when user models are built for reasoning and decision making.

The AA ontology serves as the information model of the user, and is further developed as a part of this work for the purpose to capture information obtained in an ambient environment. The results are described in Section 4.2.

In the exchange of information between systems, the concept node system of the AA ontology serves as terminology for the knowledge and reasoning nodes structured by the RKM ontology. The RKM ontology is based on the *Argument Interchange Format (AIF)* [13], expanded with nodes capturing e.g., *possibilistic values*. The information stored in the AIF-based RDF-structures is extracted into different logic languages depending on the topic and field of reasoning through an *argument and information exchange service* (Figure 1). These ontologies play a core role as a common semantic model (i.e., vocabulary) for the software agents and human actors participating in dialogues, further described in Section 5.3 and in the following section.

4.1 Data Interoperability

To have a valid vocabulary in terms of ontologies helps to manage issues of *data interoperability* between the different components of the As-A-Pal architecture. As we can observe in the As-A-Pal architecture, there are different modules which manage different sources of data. Although all of these modules have different aims, all of them provide instances of some classes of the ACKTUS core ontology. Moreover, all of them have their own ontologies which are extensions of some subsets of classes of the ACKTUS core ontology. In order to describe, how data interoperability is managed in the As-A-Pal architecture, let us denote by $\mathcal{O}_{ACKTUS}$ the ACKTUS' core ontology. $\mathcal{O}_{ACKTUS}$ provides a basic vocabulary, which is considered in the different sub-modules of the As-A-Pal architecture. In this sense, let us introduce the following ontologies:

$\mathcal{O}_{ALI}$: $\mathcal{O}_{ALI}$ is the ontology of the ALI system. This ontology contains some classes of $\mathcal{O}_{ACKTUS}$; however, it contains classes which do not belong to $\mathcal{O}_{ACKTUS}$. Let $\mathcal{O}_{ALI-nodes} = \mathcal{O}_{ACKTUS} \cap \mathcal{O}_{ALI}$.

$\mathcal{O}_{MUDRA}$: $\mathcal{O}_{MUDRA}$ is the ontology of the MUDRA system. Like $\mathcal{O}_{ALI}$, $\mathcal{O}_{MUDRA}$ contains some classes of $\mathcal{O}_{ACKTUS}$ and some other ones which do not belong to $\mathcal{O}_{ACKTUS}$. Let $\mathcal{O}_{MUDRA-nodes} = \mathcal{O}_{ACKTUS} \cap \mathcal{O}_{MUDRA}$.

$\mathcal{O}_{Kitchen}$: $\mathcal{O}_{Kitchen}$ is the ontology of the kitchen As-A-Pal. Like the previous ontologies, $\mathcal{O}_{Kitchen}$ contains classes which do not belong to $\mathcal{O}_{ACKTUS}$,

however, this ontology extends some classes of $\mathcal{O}_{ACKTUS}$. Let $\mathcal{O}_{Kitchen-nodes} = \mathcal{O}_{ACKTUS} \cap \mathcal{O}_{Kitchen}$.

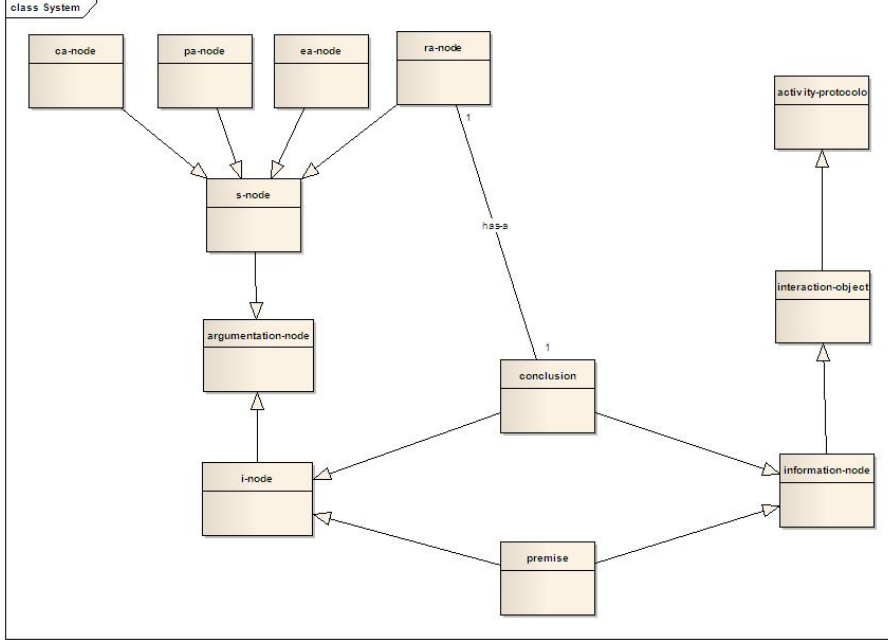


Fig. 2 Some of the core classes of the argument interchange format associated to the classes for information capture.

Informally speaking, $\mathcal{O}_{ALI-nodes}$, $\mathcal{O}_{MUDRA-nodes}$ and $\mathcal{O}_{Kitchen-nodes}$ denotes the classes which define the bridges between ACKTUS knowledge bases and the modules of the As-A-Pal architecture. Hence, the classes of $\mathcal{O}_{ALI-nodes}$, $\mathcal{O}_{MUDRA-nodes}$ and $\mathcal{O}_{Kitchen-nodes}$ play an important role in the data interoperability between the different components of the As-A-Pal architecture.

In order to manage the data interoperability in the As-A-Pal architecture, the *Argument Interchange Format* (AIF) has been adopted and extended [13]. To this end, we have defined a semantic data model based on the AIF, see Figure 2. This semantic data model defines a set of classes which capture the main concepts of the AIF, *i.e.*, *s-node* and *i-node*. Hence the semantic definition of these classes belong to $\mathcal{O}_{ACKTUS}$, $\mathcal{O}_{ALI-nodes}$, $\mathcal{O}_{MUDRA-nodes}$ and $\mathcal{O}_{Kitchen-nodes}$. Moreover, instances of *s-node* and *i-node* defines the inputs and outputs of the ALI system, the MUDRA system and the Kitchen As-A-Pal. These classes are extended in $\mathcal{O}_{ALI}$, $\mathcal{O}_{MUDRA}$ and $\mathcal{O}_{Kitchen}$ according to the data which will be shared by between the ACKTUS knowledge bases and the modules of the As-A-Pal architecture, *i.e.*, the ALI system, the MUDRA system and the Kitchen As-A-Pal.

4.2 The ACKTUS Actor and Activity Ontology as a Holistic and Generic Model of the User

Our goal is to provide methods for achieving *holistic* user modeling in smart environments. In therapeutic terms, *holistic* means to build an interpretation of a client's situation, which covers a broad range of aspects related to physical, cognitive, psychological, social factors, abilities and limitations, situated in an environment. For achieving an efficient solution, we approach the task by using the broad ACKTUS core ontology as a reference ontology and vocabulary for information about the user and a knowledge domain, while allowing the systems and system agents extracting the information, which is relevant to a situation based on what topic is in focus in a certain situation. Similarly, the different ambient systems utilize subsets of the core ontology, which are expanded in the different applications to support the more focussed purposes of the applications. As a consequence, a sub-set of the user model is instantiated in each situation, which is comprehensive and sufficient for the situation and purpose for which it is used.

The ACKTUS *actor and activity (AA) ontology* was created based on models of human occupation [25,26], the International Classification of Disability, Functioning and Health (ICF) provided by the World Health Organization⁵, and other medical terminologies [35]. The AA ontology forms the semantic model of the user, the user's activities and of the user's environment. The ontology functions as a generic core ontology for health-related domains, and is expanded with more specific information for each knowledge domain following the scope and necessary granularity of a domain.

The user of the system is represented through the AA ontology as a human who acts as a user, apart from his or her roles in e.g. a work or other social environment. As a human actor, the user has motives, interests, roles, habits, skills and knowledge in domains. Furthermore, the human has a body with functions and processes and performs activities in an environment. It should be noted, however, that we make a distinction between the user's actual e.g., performance of an activity or cognitive function, and the collected *information about* the performance of an activity or cognitive function. The information is represented as instances of i-nodes as described in the previous section. This distinction enables the treatment of conflicting or ambiguous information obtained from different sources as *defeasible* information, i.e., information which can be overruled, or defeated when new information is obtained.

As a part of this work, the AA ontology was evaluated and expanded in order to capture also environmental information obtained by a ubiquitous system.

Activity theory was used to categorize and interpret activities in order to represent knowledge about recognizable and prioritized activities [25]. Moreover, we evaluated the conceptual models underlying the two frameworks ACKTUS and *egocentric interaction* [46]. The egocentric interaction frame-

⁵ <http://www.who.int/classifications/icf/en/>

work is used in the design of the Kitchen As-A-Pal. The use case scenario of our persona Rut described in [34] was used for the purpose.

As a result, the AA ontology was extended to include the class *space*, which can be both physical and virtual. Spaces are modeled as locations where objects and actors are situated during activities (Figure 3). Features that relate to the physical home environment (located in spaces) are captured. Moreover, the conceptual model of human activity as described by Activity theory was integrated as follows. An activity is represented as a class, which has *tools* with reference to the object class, and participants referencing the actor class. This way the role of an object can be inferred by a situation, whether it plays the role of a tool or being the focus (i.e., topic) of an activity. Activities are represented by the *Activities and Participation* class, following the ICF categorization as basic structure (Figure 3).

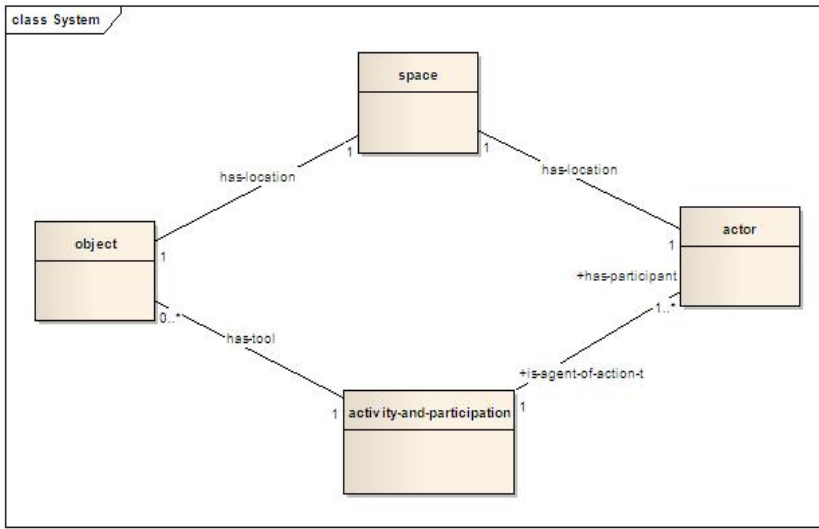


Fig. 3 An overview of the nodes in the extended core ontology that builds the user (actor), activity and tool models. The domain experts can refine the conceptual model by creating sub-classes of relevant activities, objects and spaces so that it suits the domain.

The activities that are performed by the persona *Rut* in the kitchen space during the breakfast moment of our example scenario are done partly in parallel and are the following: having breakfast (includes actions such as eating, drinking, pouring coffee into the coffee cup, and milk into the porridge plate); keeping up with society by reading news, keeping up with her social network by having a dialogue with a friend; caring for her health by taking her medication; exercise and be entertained by playing with cognitive training tool; plan the day by checking her agenda/schedule; and finally, in order to reflect upon daily life she has a weekly dialogue with the *Coach Agent* (CA) of the AAL environment (further described in Section 5.3).

Some of these activities are taking place in both a physical space (dining area) and in virtual spaces (a news forum on the web, a social media application that gives access to her friend and her home, a web-based cognitive training tool, etc) making these activities taking place in a *mixed-reality space*. This characteristic feature is also captured by the egocentric conceptual model.

A significant difference between the two models is that while the ACKTUS ontology covers purposeful activities, their tools, motives, actors and locations, in the way Rut and health professionals perceive activity, the egocentric conceptual model covers primarily concepts related to sensors, effectuators, interaction devices and their properties and their relation to the situative spaces [46]. This means that the representation focus of the egocentric interaction model lies on the *operational* level of activity in terms of Activity theory, e.g., how accessible objects are to the actor and through which modalities information is communicated. In this way the two conceptual models are supplementary. For instance, Rut's preferences regarding how to interact with the systems captured at the level of activity which can be motivated by a human, and represented by the ACKTUS semantic model, can be effectuated by the egocentric interaction framework, implemented in Kitchen As-A-Pal. An example is presented in the following section.

4.2.1 The Rut Case Example

The following example illustrates how the different levels of activity are captured in the As-A-Pal architecture. Consider the persona Rut who has been assessed having minor difficulties remembering past and future (planned) events. The assessment was based on Rut's son as a source of information. Rut herself did not perceive that memory difficulties are inferring much on her daily activities. One situation in which difficulties have been observed is when Rut needs to take the medication. The following information, represented as i-node instances of the RKM ontology, has been collected through ACKTUS applications based on interviews with Rut and her son. The associated concept from the AA ontology are shown in parentheses, which represents the purpose, or topic of the statement:

1. Heteroanamnesis shows mild memory dysfunction (memory)
2. Autoanamnesis shows no perceived memory dysfunction (memory)
3. Rut has some difficulties taking her medication (taking medication)
4. Rut thinks it is very important to take her medication (goal)
5. Rut wants support with her medication (preference)

The following information associated to the activity *taking medication* was collected through the ambient systems in the kitchen environment during one morning:

1. Rut picks the pill dispenser up from the table
2. Rut puts the pill dispenser on the table
3. Rut moves her hand towards her mouth

4. Rut picks the glass up from the table
5. Rut moves the glass towards her mouth
6. Rut puts the glass on the table

The pill dispenser is associated as a tool to the activity taking medication and Rut is the actor of the activity (Figure 3). This information is interpreted by the ambient application using the Kitchen As-A-Pal ontology, which contains the detailed information about activities at operational levels of activity. The information is used by the ambient application to infer the conclusion that this morning Rut has taken her medication with the likelihood *probably* (see more details in Section 5.2.1). This information is sent to the Actor repository using the argument and information exchange service.

This example illustrates that the ambient system collects observational information about actions at an operational level, detects a pattern, which typically leads to the assumption that a certain action is taking place. This action is further evaluated by the multi-agent systems by fusing different sources of information into an interpretation, which corresponds to Rut's goals and priorities (i.e., a higher level of activity). The management of the user model will be further described in the following section.

5 Management of the Ubiquitous User Model

The ontology-based user model was introduced in Section 4 and its role for fusion of information from multiple sources. In addition to this, As-A-Pal contains a set of functionalities and services that are combined in the creation and management of the user model:

1. Different *assessment applications* for the the health care professionals, the person in a home environment and family members (Section 5.1),
2. *Activity recognition* and updates based on the information obtained through different sensor-based systems (Section 5.2),
3. *Multi-Agent Systems* for reasoning and management of the user model (Section 5.3).

These will be described in more detail with the help of one of our personas and scenario in the following sub-sections.

5.1 Assessment Applications Developed by Domain Experts Using ACKTUS

The prototype applications developed by domain experts, which are available as part of As-A-Pal are the following:

1. DMSS-W: the web-version of a dementia management and support system, which provides reasoning and decision-making support to physicians and teams [31],
2. I-REHAB: a support application for occupational therapists, physiotherapists and nurses,

3. I-HELP / HEMMAVIS: a web-application for older adults, which contains applications supporting different aspects of daily life, and a channel for communicating with the intelligent services of As-A-Pal,
4. SAFE STEP: a behavior change system mediated through e.g., a mobile phone or touchpad for personalized support for physical exercise.

The domain experts create *assessment protocols* to be used for capturing particularities about a user instance. The assessment protocols are part of the RKM ontology, designed for providing the instruments for collecting semantic information about the user and his or her activities and environment. In this process the domain experts include specific information that is used in the tailoring of the interaction and content to an end user. The I-Rehab application was extended with a protocol for the assessment of priorities and goals as a result from an earlier evaluation study [36]. Moreover, based on the personas and scenarios, the therapists included protocols for tailoring exercise, assessing risk for falling down, pain conditions, sleep disorder and worries.

5.1.1 Capturing Motives and Goals for Determining Preferred Interventions

A new registered user of I-Help receives two assessment protocols associated to the user model, which are associated to all users and stored in their Actor repository. These are the "Identify Priorities and Setting Goals", and the "I-Help" protocols, which are used for tailoring the I-Help content. The first assessment protocol functions as a screening instrument, and collects a set of baseline information with topics common for all clients. In the case when the client describes difficulties of some kind, more detailed questions about these are given, as a way of tailoring the assessment. I-Help functions as a mashup of different services, of which some are developed using ACKTUS (e.g., Safe Steps).

Our Rut persona participates in a baseline assessment, which is conducted together with the therapist guided by the assessment protocol "Identify Priorities and Setting Goals". A clinical interview takes place, which primarily aims at *seeking information* in order to form a base for intervention.

The assessment and its outcome is represented in the Actor repository as an instance of the *investigation-event* class, which has a set of *evidence* in the form of references to i-nodes located in the RKM ontology. An i-node in turn, has a *concept* property, which situates the information in the AA ontology and its medical and health terminology. Moreover, the execution of the assessment protocol is logged, for the purpose to evaluate the dialogue lines. i-nodes stored in the Actor repository may also represent *advices* given or *encouraging arguments* to be given, or assessment protocols (i.e., a kind of *activity*), which the user can access through the I-Help application.

A subset of the collected information, which is specific for Rut's situation is shown in Table 2, where the associated concepts are shown within parentheses. The priority of each goal is set according to an ordered list of values. Also the level of satisfaction is measured according to an ordered list of values, obtained

Table 2 Rut’s priorities and goals.

Priority level	Prioritized Topics	Valuation of Goals	Preferred support
1	Rut is <i>very</i> worried (worries)	Rut thinks reducing worries is of <i>highest</i> importance	Yes
1	Rut is worried about falling down (fall)	Rut thinks reducing worries is of <i>highest</i> importance	Yes
1	Rut is seeing close friends and family <i>too seldom</i> (keeping up with friends and family)	Rut thinks meeting friends and family is of <i>highest</i> importance	Yes
2	Rut has <i>some difficulties</i> taking her medication (taking medication)	Rut thinks it is <i>very important</i> to take her medication	Yes
2	Rut’s pain is <i>moderate</i> (pain)	Rut thinks it is <i>important</i> to reduce her pain	Yes
3	Rut thinks that she <i>is able to</i> keep up with society to the extent she wants (keeping up with society)	Rut thinks that it is <i>important</i> to keep up with society	No
3	Rut sleeps <i>well</i> with the help from her sleeping pills (sleep)	Rut thinks it is <i>important</i> to be satisfied with her sleep routines	Yes
4	Rut has <i>some difficulties</i> preparing meals (kitchen tasks)	Rut thinks that it <i>is not that important</i> to have at least one cooked meal each day	No

from the assessment instrument on which the assessment is based. The two can be weighted together by the system agents, based on the principle that the activities with the highest priorities in combination with lowest satisfaction are ranked with highest priority. A priority list based on Rut’s situation is shown in Table 2.

Based on the information and the priorities which can be inferred, a set of interventions are decided to be implemented for improving Rut’s situation. In our scenario the interventions come in the form of ambient systems, which monitor daily activities related to Rut’s goals, and which provide the tailored feedback.

A set of prioritized assessment activities are associated to the user’s profile for following up on improvements or need for changes in the interventions. These activities are represented as assessment protocols, which are used for continuous followup assessments. In Rut’s case the associated assessment protocols are ”Risk for falling” and ”Pain assessment”. Moreover, the therapist and her client Rut decide together what will be followed up regularly, e.g., once a week. A set of key questions are identified, and becomes the content of a tailored followup assessment protocol, which is also stored in the Actor repository.

5.2 Integrating Information Obtained from Ambient Systems

Our goal is to develop a ubiquitous user modeling infrastructure that is centered on the human's resources such as body, skills and mind, and which frames the dynamic environment with all its inherent complexities from the human agents perspective. To this end, we integrate different models for capturing ambient information, and for refining the information. We develop methods for the following three approaches:

1. A *sensor network* integrated in a kitchen environment (Kitchen As-A-Pal), which detects state transitions for common objects, human presence, and object manipulations [47, 40],
2. *3D sensor-based*, which is a lightweight solution for detecting body posture, gestures, objects and object manipulations (MUDRA) [6],
3. *Mobile device and its sensors*, to be carried around in and outside a home and/or work environment, detecting different types of movements and activity patterns (ALI) [18, 19].

The data collected from the different sources are pre-processed for obtaining knowledge about the actor's potential activities and activity patterns, which can be communicated using the common vocabulary. This is done in the three software modules the Kitchen As-A-Pal, MUDRA and ALI respectively (Figure 1). These represent sources of information, which are used for fusing the information and validating interpretations of the information about human activity using the ontologies introduced in Section 4.1.

In the following subsections we describe in more detail how the information obtained by an ambient environment contributes to the user model, the assessments and planning of interventions and followups. The personas and scenarios serve as running examples.

5.2.1 Morning Activities in a Smart Kitchen

Kitchen As-A-Pal is an interactive smart kitchen where a human actor can conduct the typical kitchen activities according to their routines, and receive personalized support for compensating for decreased abilities. The kitchen constitutes a building block for our holistic approach to ambient assisted living and serves as an environment for developing and testing ambient and smart technologies. The sensor infrastructure of kitchen As-A-Pal provides information about the occupants of the kitchen including their location and interaction with kitchen objects. Information from the sensors are used by the *egocentric middleware* for recognizing human situation and activities [40], and for offering interaction management support [47].

Action specification languages are being explored where an activity is formally modeled as a sequence of actions [40]. The Language $\mathcal{A}$ described by Gelfond and coworkers in [16] is used as a starting point and is extended [4]. A *fluent* represents a property of an object and a *state* represents a collection of fluents. In this setting, a set of fluents, which describe the objects which

are present in Kitchen As-A-Pal, gives a state of the Kitchen As-A-Pal. In other words, a state describes a *picture* of the Kitchen As-A-Pal. A state of the world (in our case the Kitchen As-A-Pal) is modified by the execution of actions. This means that, at least, a property (a fluent) of an object of the world is changed by the execution of an action.

A *situation* is described as a history of executed *actions*. In the initial situation no action has been executed: we represent this by the empty list []. The situation $[a_n, \dots, a_1]$ corresponds to the history where action a_1 is executed in the initial situation followed by a_2 , and so on until a_n . The extended language $\mathcal{A}$ is used for formally describing *actions* which define human activities. Situations are inferred from the declarative specification of actions and the observations from Kitchen As-A-Pal. Activities are modeled using *trajectories* [39]. A trajectory is basically a *meta-library of a sequence of actions*. The trajectory $[a_m, \dots, a_1]$ captures an activity which is composed by m actions. Like a situation, a_1 is the initial action which is followed by action a_2 , and so on until a_m . In this setting, different trajectories are defined a priori in order to detect activities, which potentially were performed by an user. In this sense, by matching *situations*, which were inferred by the observations from the Kitchen As-A-Pal, and *trajectories*, different activities can be detected. Since this matching can be partial, which means only part of a trajectory appears in a situation, different *likelihoods of uncertainty* with respect to the performance of activities are defined. This situation is illustrated in the Rut scenario.

One of the prioritized activities in the Rut scenario is taking medication (Table 2). In order to define a trajectory the activity taking medication, let us consider the following actions exemplified in Section 4.2.1, and which represent how Rut typically performs the activity:

Action	Meaning
a_1	Rut picks the pill dispenser up from the table.
a_2	Rut puts the pill dispenser on the table.
a_3	Rut moves her hand towards her mouth.
a_4	Rut picks the glass up from the table.
a_5	Rut moves the glass towards her mouth.
a_6	Rut puts the glass on the table.

Before moving on, we observe that for inferring these actions, the Kitchen As-A-Pal has been provided with a 3D-sensor; moreover, the objects such as glasses has been augmented with different types of sensors including temperature, humidity, force, infrared distance, light and RFID readers capable of delivering streams of data. These streams of data have been associated to a finite set of fluents with respect to the objects and the user of the the Kitchen As-A-Pal.

The following trajectory of actions suggests that Rut takes her medication the way she typically does:

$$T_{\text{taking medication}} := [a_6, a_5, a_4, a_3, a_2, a_1]$$

Let us suppose that one morning the system infers the following situation based on the observations obtained by the Kitchen As-A-Pal:

$$S_{morning} := [a_3, a_2, a_1]$$

By matching the actions of $T_{taking\ medication}$ with the action of $S_{morning}$, we can observe that there are actions of the trajectory $T_{taking\ medication}$ which are missing at this occasion, when comparing this event to the example presented in Section 4.2.1 and to the trajectory. Therefore, the likelihood that she has actually taken the pills is *possibly*, and not *probably*, and represents consequently a weaker assessment than in previous example⁶. This information is represented as an instance of an i-node with an association to one of the common classes of the ACKTUS AA ontology. The information is sent to the Actor repository using the argument and information exchange service.

5.2.2 Activity Patterns Detected by a Mobile Device

ALI [18, 19] is an alternative for tracking behavior such as position and movement of a person, analyze the data and respond to the user in a relevant and useful way by making use of the technical characteristics of a mobile phone. The purpose of ALI is to empower the individual, offering recommendations to change habits of persons and stimulating changes of behavior through positive feedback for her/his actions.

ALI represents a data source, which feeds information in the form of i-nodes to the Actor repository in a similar way as the Kitchen As-A-Pal. Moreover, ALI can retrieve information from the ACKTUS repositories, for reasoning about activities over time, e.g., explaining and justifying activities of individuals. ALI integrates a possibilistic argumentation-based decision making framework [38] and Activity Theory [29] in order to obtain argumentative explanations for the individual's behavior, and in this way deal with uncertain and incomplete information obtained from observations of the individual's context. ALI obtains data from the embedded sensors in the mobile phone, processes contextual information, which partly is obtained from ACKTUS and generates argument-based recommendations which are displayed to the individual. Recommendations and arguments are stored in order to monitor a potential behavior change over a period of time.

In the Rut scenario, the following activities were prioritized and selected to be supported: reducing worries, e.g., for falling down, taking medication, reducing pain conditions, and maintaining satisfactory sleep routines (Table 2). Therefore, we focus on the sleep pattern analysis and exercise routine analysis services supported by ALI. Sensor data is obtained by the position and accelerometer sensors, which are filtered and refined by the ALI application into information represented as instances of i-nodes in the RKM ontology, which can be sent to the Actor repository. In the Rut scenario, Rut begins

⁶ The likelihoods are defined by considering *fuzzy distributions* in order to manage linguistic labels, e.g., possibly, probably.

to walk to the bathroom during night time after a period of time. Since this represents a change of behavior, the new information about Rut’s sleep and activity pattern (Rut is walking around nighttime with the likelihood *probably*) comes into conflict with the old information given by ALI that Rut typically sleeps the whole night.

The same type of information can be used in the cases of the Eva and Thea personas. For these two personas the important aspect is the day and night rhythm, balancing rest and activity, for them to be able to perform in a satisfactory way during the hours they need to be productive.

How insufficient or conflicting information, which reaches the Actor repository, is handled is further described in the following section.

5.3 Agent-Based Management of the User Model

The motivation for using multi-agent dialogue systems in our work is that this approach allows for modeling the conditions for reasoning and knowledge generation, in spite of ambiguous, uncertain and incomplete domain knowledge and knowledge about the user’s situation. The agency of semi-autonomous software agents can be used for mixed-initiative collaborative reasoning and actions, involving also human agents. In addition, we anticipate that agent-based dialogues will facilitate the interaction design of more intuitive and natural dialogues between the user and the system.

However, in the current implementation the knowledge and content of dialogues are restricted to the content of the ACKTUS repositories, which puts limitations to the ”naturalness” of dialogues.

At the core of the intelligent services provided the actor is the *Coach Agent* (CA) (Figure 1), which manages the user model by reasoning with other agents about conflicting interpretations of the user’s activities, and preferences obtained from different sources [37].

When executing the structured *information-seeking dialogues* guided by the assessment protocols, information about the client, or *actor* is generated (e.g., Table 2). These protocols are executed during the clinical interviews taking place between the therapist and the client. Based on this information stored in the Actor repository, the CA forms a baseline model of the actor based on the AA ontology, identifies prioritized goals, what activities are prioritized, and in what situations the system is expected to act.

Based on the Rut scenario, and the situations described in the previous sections, the CA has the following information. Firstly, the following activities were prioritized and selected to be supported: reducing worries, e.g., for falling down, taking medication, reducing pain conditions, and maintaining satisfactory sleep routines (Table 2). Secondly, the CA receives insufficient information about taking medication from the Kitchen As-A-Pal, and conflicting information about sleep routines received from ALI. Consequently, the CA needs to act upon these situations. Following the scenario, the CA has different possi-

bilities to act. The CA can initiate a dialogue with other software agents to resolve contradictions, or with the user Rut for the same purpose.

The purpose of the software agents' dialogues is to share (defeasible) knowledge to construct arguments supporting beliefs, and in this way increase each agent's knowledge repository (belief base). The dialogues can be activated by any one of the agents (the human agent through the I-Help application), and the content of messages is selected from their belief bases, based on the *topic* for the dialogue [30]. The topic is framed by the underlying domain ontology developed by the domain experts, which defines the reasoning context and associated critical questions, rules, knowledge sources, etc. [35,33]. Critical questions, their typically structured answer alternatives and the text labels of i-nodes retrieved from the ontology function as the content of locutions in dialogues. In this way, the human agent is able to follow the agents' reasoning and also participate in the dialogues.

In the current implementation of the multi-agent system, which handles dialogues with the human agent, the CA uses the Actor repository as a source for information. The CA retrieves information about the user, which is related to the current topic, or topics. Based on this information, the CA creates a user model, which is enhanced or adjusted through the dialogues with the human actor or other software agents, e.g., the domain agent (DA). The DA retrieves knowledge from a particular domain knowledge repository, also based on the current topic, or topics. The outcomes of the dialogues in terms of new knowledge, is stored in the Actor repository.

5.3.1 MAS Example 1: Resolving a Situation with Insufficient Information

When the CA assesses the morning scenario, it has insufficient information about whether Rut has taken her medication this morning since the likelihood *possibly* is too weak to be sufficient. In this situation, the coach agent initiates a dialogue with Rut by asking if she has taken her medication this morning. The following is an example of a possible dialogue line:

- CA: *Have you taken your medication today?*
- Rut: *I don't know...*
- CA: *Are the pills for this morning still in the pill dispenser?*
- Rut: *No.*

5.3.2 MAS Example 2: Resolving a Situation with Conflicting Information

When the information is obtained from ALI that Rut is walking around during nighttime, the CA finds that this contradicts the information obtained from Rut that she sleeps well with the help of her sleeping pills (see Table 2) and earlier information from ALI. The CA initiates a dialogue with the *Domain Agent* (DA) representing the ACKTUS-Rehab domain knowledge, with the topic *probably walking around nighttime* for the purpose to build and evaluate

arguments related to the topic. Apart from the argument built upon the information from ALI, there are also arguments, which can be built based upon Rut's own information and the ACKTUS-Rehab knowledge base.

An argument is understood as a statement about a topic. In our approach we consider three binary relationships between arguments: *attack*, *undercut* and *support*. Both attack and undercut are understood as disagreement between arguments and support is understood as an agreement between arguments. An *argumentation tree* usually is formed by a set of arguments and one of them is regarded as the root. The rest of arguments are related to the root by one of the binary relations. Informally speaking, we can say that an argument is a winner in an argumentation tree if there is a set of arguments S which supports it and all the argument which belong to S are also winner. Any argument which has no attackers (arguments which disagree with it) is considered a winner argument.

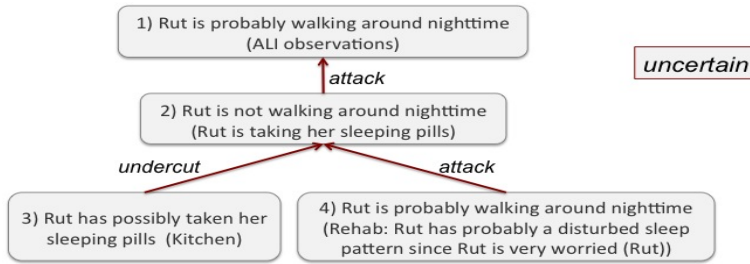


Fig. 4 Arguments and argumentation tree without a winning argument.

The following are examples of arguments, showing the source of the knowledge in parentheses (Figure 4):

1. Rut is probably walking around nighttime (ALI)
2. Rut is probably walking around nighttime, based upon: Rut has probably a disturbed sleep pattern since she is very worried (Rehab)
3. Rut is not walking around night time since she is taking her sleeping pills (Rut)
4. Rut has *possibly* taken her sleeping pills (Kitchen As-A-Pal)

In this situation the weak support for the argument that Rut has actually taken her medication obtained from the Kitchen As-A-Pal, is weakening the argument that Rut is not walking around nighttime. Consequently, the stronger argument is that Rut is walking around nighttime, which leads to the knowledge that this may be due to her worries or because she may not have taken her medication.

CA needs to act upon information related to the prioritized topics (Table 2) and now the following are actualized: *sleep*, *taking medication* and *worries*, and in particular, worries about falling down. Following the priority levels, CA

initiates a new dialogue with the DA about whether there is an increased risk for falling, which is also confirmed by the DA based upon the information.

At this point, the CA has information, which is in conflict with the goals prioritized by Rut. The CA now has the option to initiate a dialogue with Rut in order to verify the information. Since there is an assessment protocol designed for following up on Rut's interventions with respect to her priorities, in a weekly routine, the CA uses this. It contains questions about amount of worries, quality of sleep and medication. The CA receives updated information from Rut, which is in our example the following:

1. Rut is worried about falling (worries)
2. Rut did not sleep well the past night (sleep)
3. Rut has taken her medication (taking medication)
4. Rut thinks that the reason for not sleeping well the past night is due to her pain (pain)

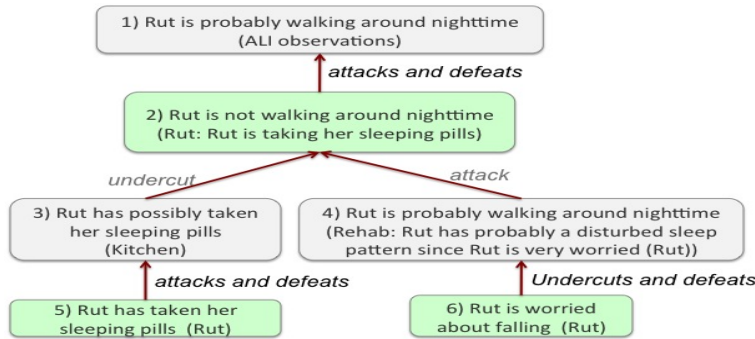


Fig. 5 Arguments and argumentation tree with the undefeated "winning" arguments 2, 5 and 6 shown in green color.

Compared to earlier assessments, Rut has now less worries, which decreases the likelihood that she is having a disturbed sleep pattern, and consequently, less likely wandering around nighttime. However, she now does think that she is not sleeping well and that this is because of her pain condition (Figures 5 and 6). The CA can conclude that there is an increased risk for falling, due to the increased pain condition, and due to that the sleeping pills does not lead to good sleep, both causing the walking around night time. The CA has two options to act upon the information. The CA can do nothing and await new information about coming nights from ALI and Rut, or ask Rut whether she wants to discuss her medication and pain condition with the nurse.

The examples show the sources of information, which can be both systems and human actors. The multi-agent systems integrate a differentiation between measures of the reliabilities of sources. In the case of sources representing human actors and their knowledge, then the knowledge of an *expert* weights more than an *actor who is in a position to know*, however less compared to

established *evidence-based knowledge* or *best practice guidelines*. In this work, the different ambient systems still have the same weight as sources, however weaker than the actor or expert. Depending on how reliable the information obtained by these systems are evaluated to be in user studies in the future, this weight will become adjusted accordingly.

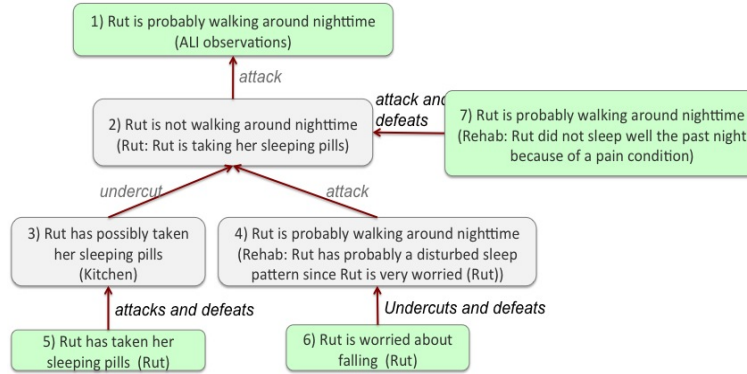


Fig. 6 Arguments and argumentation tree with the undefeated "winning" arguments 1, 5, 6 and 7 shown in green color.

6 Evaluation Study

The following research questions were identified in the introduction, and an implementation of a solution was presented in Section 5.1.1:

- How does the user model capture motivational structures?
- How is intervention defined based on the initial assessments?

We conducted a pilot case study targeting the initial phases of the therapeutic assessment and intervention procedure: initial assessment, the establishment of a holistic understanding of the client (i.e., the *user model*), a plan of interventions is made, implemented and an initial followup on the implementation is done. However, studying the effects during a longer period of time, enabling followup assessments, which lead to adjustments of both the client model and interventions according to detected changes in needs, ability, health and preferences, we leave for future work.

As a part of this work, we are interested in combining the initial assessments with the ambient systems in the environment in order to address the following research question:

- How is the information obtained from the ambient environment contributing to the user model, initial assessments, the planning and implementation of interventions and followup assessments?

In addition to these questions, we are interested in exploring how the participants approach the task of defining own arguments, which may increase the actor's compliance with suggestions for changed behavior.

6.1 Participants

We recognize that the conditions, which the personas presented in Section 6.1 represent are common, triggered by circumstances in the social and cultural context and may affect anyone at some point in their lives (e.g., [1]). Therefore, we selected a convenient sample of three subjects for our study who were not necessarily suffering from any of these conditions, but who matched the age categories young adults (18-24) or older adults (older than 65 years). The participants were two young female adults (18 years old), who we call Anna and Sara, and an older female adult (67 years old), who we call Sabina. Written informed consent was obtained by the participants.

In addition to these three participants, two occupational therapists were participating for conducting the assessments and forming plans for interventions. The two therapists also participated in the knowledge engineering of the ACKTUS-Rehab application. Additional four physiotherapists participated in the knowledge engineering of the Safe Step application.

6.2 Methods and Procedure

Our study was organized in the following way, following the typical routine by which rehabilitative interventions are formed, deployed and followed up.

Each of the three participants was at an initial occasion interviewed and assessed by an occupational therapist. In this process they used the I-Rehab application. In the case of the older adult Sabina, the protocol for the *Assessment of Goals and Priorities* was used, together with a structured and guided assessment of *Risk for Falling Down* due to the worries expressed about this by Sabina.

In the case of the two young adults, the protocol for the *Assessment of Goals and Priorities* was used in addition to the *Safe Step assessment*. In one of the young participants the protocol for *Sleep Disturbance Assessment* was also used, since the importance of resolving this was assessed with the highest priority by the participant. In all cases the client used a laptop computer while providing the information.

Depending upon the participant's needs, goals and priorities, activities considered as important to the individual were identified by the individual and the therapist together, which were to be supported. A corresponding assessment was done by the Coach Agent (CA) based on the information as baseline for the user model.

At a second occasion the participants were observed and assessed, while conducting one or more selected and prioritized activities that they wanted

to be able to accomplish. These two assessments formed a baseline for the interventions and for the As-A-Pal systems. At the second occasion selected functionalities of the smart environment was applied, partly to form a baseline also for the system. The therapist and the participant formed based on the assessments a plan containing what interventions would be suitable to implement for the purpose to improve their situation.

In the second assessment of the older adult the activity was a morning routine where Sabina prepared breakfast according to her routines and had breakfast while reading the news and socializing with a friend through a social network application. The information collected and the assessment done during the morning routine was used as a baseline for the activities that Sabina had prioritized to become supported by the support system. The activities of the scenario were conducted in a controlled smart kitchen environment partially shown in Figure 7.



Fig. 7 The As-A-Pal kitchen environment.

The two young participants used the mobile ALI support application during a short initial period of two days in order to collect observational information to be used for forming a baseline assessment in their cases. After the two days a follow-up assessment was conducted with the therapist, which resulted in an adjusted plan for interventions.

For tailoring the support for the targeted activities and addressing the fourth research question, the two young adult participants modeled personalized recommendations which they preferred to be given, and under what conditions they should be presented. They used the information/advice-editor and the rule editor of ACKTUS-Rehab to do this task. These recommendations were integrated into the system before they used ALI in the baseline assessment.

6.3 Results

The results address how the user model is created and defined in the three use cases in order to achieve tailoring of services. We limit our focus and descriptions of the content of assessments and user models, to how the user can define and prioritize their motives and goals for change, in a way so that the system can capture, integrate and utilize motivational structures in the user model.

The results of the evaluation study is organized in the following subsections into results regarding the initial assessment, the creation and management of a user model in the young adults and in the older adult Sabina's case, and in what way the information obtained from the ambient systems in the environment contributes to the user model, assessments and the planning and implementation of interventions.

6.3.1 Initial Assessment of the young Participants and the Creation and Management of their User Model

The two young adults had different priorities and goals. Sara assessed highest importance to all social activities, and wanted support for finding and maintaining a healthy level of activity and exercise routines (Table 3). Anna rated generally the importance of social activities low, even grading being in public contexts as occurring far too often (Table 4). Anna wanted support for maintaining a healthy sleep routine, medication routines, and was aware that she needed support for routines regarding school and work. However, to receive support for this was not prioritized by Anna, as can be seen in Table 4.

Based on these assessments, the selected activities were further analyzed in more detail in order to design the support, and the participants' own designed arguments for encouraging desired activities were put into a context. Based on the initial assessment, the therapist selected significant questions related to the individual and created a tailored followed up dialogue protocol, which the CA uses in the regularly follow-up dialogues with the client. Figure 8 shows the summary of a follow-up dialogue with Anna. This follow-up assessment led to a revision of the user model (Table 5), and the priority list in Table 4 since the sleep had become disturbed due to a severe pain condition.

Based on the initial assessments, the following plans for interventions were created for the two participants.

Sara's interventions targeted the rhythm and amount of activities, with emphasis on physical exercise as a first step (Table 3). She received a cellphone with a tailored ALI application, which she used during a second assessment period for forming the intervention so that it would suite her daily life. A followup assessment was done by the therapist after a few days, with the purpose to evaluate and adjust the application.

The advices provided Sara were perceived appropriate, but not timely, which was due to that the system was unaware of what activities Sara had

Table 3 Sara's priorities and goals.

Priority level	Prioritized Topics	Valuation of Goals	Preferred support
1	Sara is exercising <i>too little extent</i> (exercise)	Sara thinks it is <i>most important</i> to maintain exercise routines	Yes
1	Sara is seeing close friends and family <i>too seldom</i> (keeping up with friends and family)	Sara thinks meeting friends and family is <i>most important</i>	Yes
2	Sara is able to begin with the things she needs to do to <i>a bit too little extent</i> (duties)	Sara thinks it is <i>very important</i> to accomplish her duties in a satisfactory way	Yes
2	Sara is participating in social activities <i>to the extent she wishes</i> (keeping up with friends and family)	Sara thinks that being in environments among a lot of people is <i>most important</i>	No
4	Sara thinks that she keeps up with society to a bit too little extent (keeping up with society)	Sara thinks that it is <i>not important</i> to keep up with society	No
4	Sara has <i>no difficulties</i> preparing meals (kitchen tasks)	Sara thinks that it <i>not important</i> to have at least one cooked meal each day	No

ongoing in different situations other than those that the system was designed to recognize.

A technical obstacle for Sara was the limited power of the mobile device for having the ALI application running, and remembering to bring the charger to e.g., friends or the summer house. This, however, was not a problem for Anna, since she rarely left her home for longer time periods than an hour.

Anna's interventions targeted work, meal, exercise and sleep routines, with reminders created by Anna particularly about medication, exercise and visiting her grandmother (Table 4). She also received a cellphone with a tailored ALI application, which she used during a second assessment period of four days for collecting information which could be used for forming the intervention so that it would suite her daily life. The application tracked the activity and resting/sleep routines which she currently had, which was analyzed and followed up twice by the therapist during the test period together with Anna. Based on the first followup analysis, Anna decided what would be a first step towards reaching her goals, which the application could support. At the time of the first followup, the daily routines were affected by an authentic pain condition, and the current main goal at that point was to decrease the pain and worries, which had increased due to the pain condition (Table 5). What was evident in her case was that the initial assessment was too general for her situation, visible in particular in that the scale alternatives for assessing pain and worries seemed to be too few. As a result, the therapist included a tailored assessment functionality in her profile which utilized a visual analogue scale

Table 4 Anna's priorities and goals at the baseline assessment.

Priority level	Prioritized Topics	Valuation of Goals	Preferred support
1	Anna has <i>some difficulties</i> taking her medication (taking medication)	Anna thinks it is <i>important</i> to take her medication	Yes
1	Anna is <i>pretty</i> worried about what people think (social) and about becoming ill (sickness)	Anna thinks reducing worries is <i>important</i>	Yes
1	Anna is exercising <i>rarely</i> (exercise)	Anna thinks it is <i>important</i> to maintain exercise routines	Yes
2	Anna sleeps <i>well</i> (sleep)	Anna thinks it is <i>most important</i> to be satisfied with her sleep routines	Yes
2	Anna is able to begin with the things she needs to do to <i>a bit too little extent</i> (duties)	Anna thinks it is <i>important</i> to accomplish her duties in a satisfactory way	No
2	Anna is seeing close friends and family <i>enough</i> (keeping up with friends and family)	Anna thinks meeting friends and family is <i>important</i>	Yes
3	Anna thinks that she keeps up with society to a bit too much extent (keeping up with society)	Anna thinks that it is <i>important</i> to keep up with society	No
4	Anna is participating in social activities <i>far too much extent</i> (keeping up with friends and family)	Anna thinks that being in environments among a lot of people is <i>not important</i>	No
4	Anna has <i>no difficulties</i> preparing meals (kitchen tasks)	Anna thinks that it <i>very important</i> to have at least one cooked meal each day	No

(VAS), containing 11 severity levels (results from using this is shown in Figure 8). The observation was made by the second followup occasion that when this was applied, she did not perform activities when the levels of pain or illness were 6-8, instead she went to sleep. After sleeping, the pain and illness feeling had decreased to a manageable level, 2-4, and in this situation she was also able to perform the activities she wanted to do. A conclusion that was drawn was that the CA should tailor the support according to the current state of a person, which can vary significantly between hours or days. Consequently, methods are needed to detect the state as often as necessary, tailor the support accordingly and mediate the support in a purposeful way.

The advices provided Anna were also perceived appropriate, and also timely. Timely in the sense that she could have complied with the advices, if she had been "a bit more motivated". When Anna was shown a summary of her activities detected during the days she had been using the ALI application, her interest became triggered. The reason was likely because she received an overview of her accomplishments in the form of walks through the forests with

Question (Swedish)	Answer
How would you like to say that your health is today?	Pretty good
Do you take any of the following medication?	Antidepressive
In case you use medication, how do you think it works for you to administer and take your medications?	3 some difficulties
Is there something that you have been worrying about the past days?	To become ill
How important is it to you to find ways to reduce your worries?	For what others think
Do you have pain?	Most important
How strong is your pain right now?	Yes
How strong is your sickness feeling right now?	8
Do you have difficulties falling asleep?	8
Why did you not sleep well last night?	Yes
How do you manage to use public transportation such as bus, train, flight?	Pain
To what extent are you getting into the exercise routines?	1 too difficult, impossible
How many days have you been out of your house this week?	a bit too little extent
To what extent are you able to begin with doing things that you need to do?	four of the days
To what extent are you able to meet the most significant persons that you want to socialise with?	a bit too little extent
To what extent are you able to meet other persons in your social network that you wish to socialise with?	about the extent that I wish
To what extent do you find yourself participating in social event (where known and unknown persons are, e.g., public meetings, theater)?	about the extent that I wish
To what extent do you have a good/healthy proportion of activities ongoing at the moment?	far too much extent
	a bit too much extent

Fig. 8 A summary of the follow-up assessment of Anna.

Table 5 Anna's revised priorities and goals after the first followup assessment.

Priority level	Prioritized Topics	Valuation of Goals	Preferred support
1	Anna is <i>very</i> worried about what people think (social) and about becoming ill (sickness)	Anna thinks reducing worries is <i>most important</i>	Yes
1	Anna sleeps <i>badly</i> (sleep)	Anna thinks it is <i>most important</i> to be satisfied with her sleep routines	Yes
2	Anna has <i>some difficulties</i> taking her medication (taking medication)	Anna thinks it is <i>important</i> to take her medication	Yes
2	Anna is exercising <i>rarely</i> (exercise)	Anna thinks it is <i>important</i> to maintain exercise routines	Yes

her dog and the maps were familiar. However, since the evaluation study did not continue during a longer period of time, the potential impact such feedback could have on her motivation remains to be investigated. The presentation of selected parts of the user model in relevant situations may contribute to motivating the user to comply with the chosen activity support, which will be investigated in future studies. An overview of the assessment of Anna is shown in Figure 9.

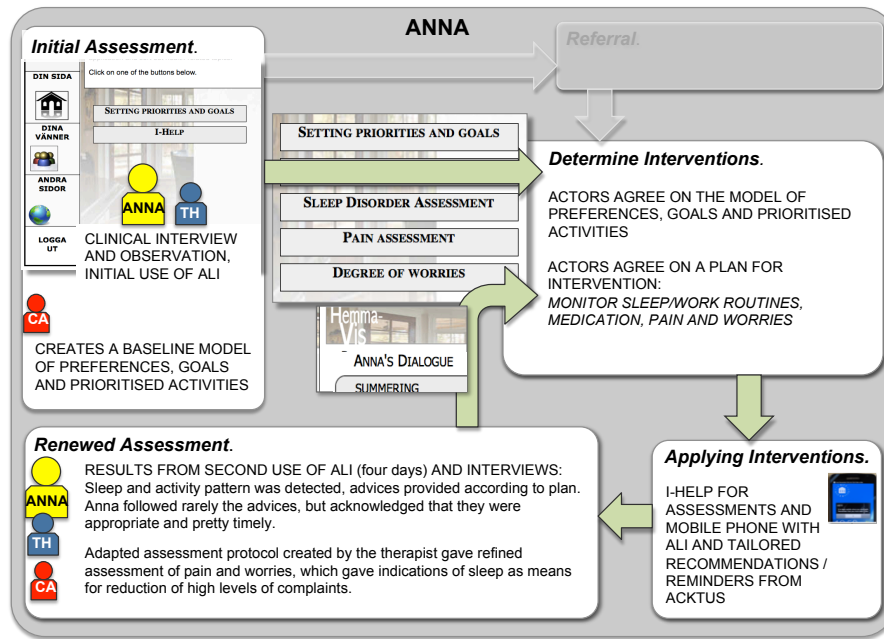


Fig. 9 Overview of the assessment in the Anna use case.

6.3.2 Initial Assessment of the Older Adult Participant and the Creation and Management of her User Model

The resulting user model of the older adult Sabina included information such as that she has a mild subjective memory impairment, she has walking ability with the support from a walker, and a visual impairment, which had occurred due to a disease affecting the eyes. Sabina's priorities are summarized in Table 6. Sabina was, apart from the clinical interview, assessed in the controlled kitchen environment (Figure 7) while preparing and having breakfast. The therapist observed the activity, assessing motor and process skills.

Sabina had done part of the breakfast routine in the controlled environment a few days before the assessment, then for the purpose to explore which equipment and procedure was familiar to her, and for preparing the assessment. At this occasion an assessment was also done to inform the ambient system about the *perceivable* spaces, due to her visual impairment, and the *reachable* spaces, due to her need to use the walker for stability. This information would have been used at a later stage in case the study had included a longer period of testing. Then feedback and information would be given to Sabina about the environment, e.g., if she has lost some object due to her sight problem.

During the assessment, Sabina was moving around in the kitchen using her walker, which she also used as a tray for the objects, which she wanted to carry along, e.g., from the fridge to the preparation space, and to the kitchen

Table 6 Sabina's priorities and goals.

Priority level	Prioritized Topics	Valuation of Goals	Preferred support
1	Sabina is <i>very</i> worried about falling down indoors and outdoors (fall), to become ill and not get help when she needs	Sabina thinks reducing worries is of <i>highest</i> importance	Yes
1	Sabina is seeing close friends and family <i>too seldom</i> (keeping up with friends and family)	Sabina thinks meeting friends and family is of <i>highest</i> importance	Yes
2	Sabina has <i>some difficulties</i> taking her medication (taking medication)	Sabina thinks it is <i>very important</i> to take her medication	Yes
2	Sabina has <i>some difficulties</i> preparing meals (kitchen tasks)	Sabina thinks that it <i>very important</i> to have at least one cooked meal each day	Yes
2	Sabina is exercising <i>rarely</i> (exercise)	Sabina thinks it is <i>very important</i> to maintain exercise routines	Yes
2	Sabina <i>rarely</i> sleeps a whole night (sleep)	Sabina thinks it is <i>very important</i> to be satisfied with her sleep routines	Yes
2	Sabina's pain is <i>moderate</i> (pain)	Sabina thinks it is <i>important</i> to reduce her pain	Yes
3	Sabina thinks that she <i>is able</i> to keep up with society to the extent she wants (keeping up with society)	Sabina thinks that it is <i>important</i> to keep up with society	No

table. She prepared porridge, boiled an egg on the stove, cooked the porridge in the microwave oven and toasted a piece of bread in the toaster after having seated by the table. While sitting by the table she read the news and had her porridge.

The therapist observed that Sabina was able to carry out the breakfast routine as usual, organizing her things and procedure in an sufficiently efficient way. However, the mild memory dysfunction was observable in the forgetting of the coffee and pills (the therapist prompted Sabina about this), and a slow pace was also observed in thinking and acting. The observation confirmed the interview-based assessment done earlier.

The As-A-Pal kitchen system detected that activity was performed, which involved the preparation space and the kitchen table space, the same way as was described and evaluated in [45]. The 3D-sensor-based MUDRA system collected information about Sabina's actions, such as moving the coffee pot to the coffee cup's location, and back to it's original location, moving the cup towards her mouth, her hand reaching for the medication located in the nearby window, moving the pill dispenser to the table location, and bringing something to her mouth afterwards. These observations could be interpreted

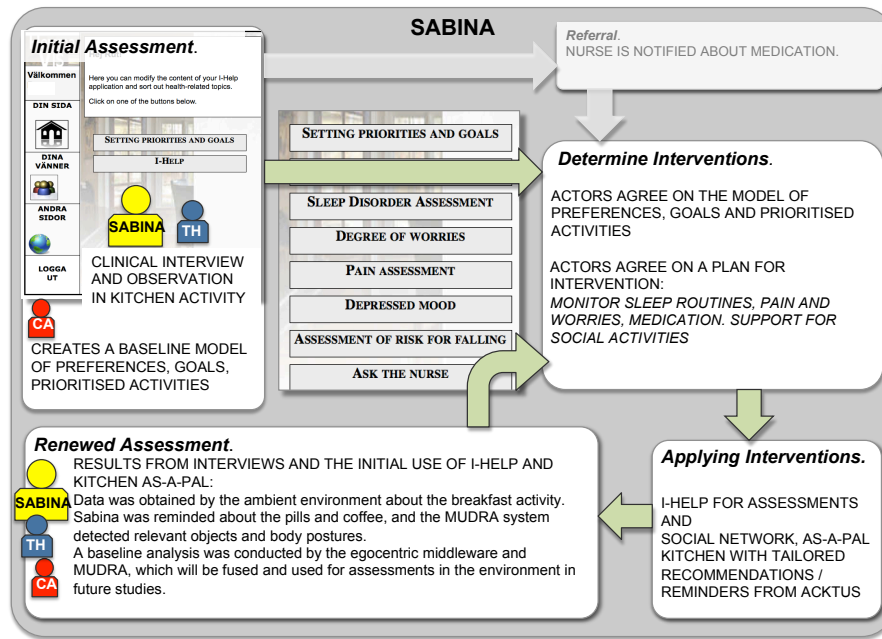


Fig. 10 Overview of the assessment in the Sabina use case.

as Sabina has had coffee as part of her breakfast, and that Sabina had taken her medication with the confidence degree *probably*.

Based on the initial assessments the following intervention plan was created. Sabina needed and desired support for managing medication, reminders about meals and social contacts, and for reducing worries, isolation and pain (Table 6). Sabina would have access to the ambient environment containing the smart kitchen, the MUDRA system, I-Help and the associated services which Sabina had prioritized. An overview of the assessment of Sabina is shown in Figure 10.

6.4 Summary

The initial assessment was done in a two-step procedure, with an interview-based assessment, supplemented with an observation-based assessment when the client is performing a chosen activity, or activities, possibly with the help of assistive technology. The observation-based assessment was done by therapists partly with the same purpose as the sensor-based ambient technology integrated in As-A-Pal, which is detecting and analyzing abilities and disabilities when performing activity. Both the therapist and systems build a baseline model of the user's activity performance, which is aimed to be used when evaluating interventions. Consequently, both benefitted from the different assessments. For instance, in Anna's case when determining suitable interventions,

Anna's subjective view of her routines was supplemented with the objective information obtained during the period of using ALI. ALI and Kitchen As-A-Pal could apply Anna's and Sabina's priorities of activities for the purpose to reducing the amount of data to be refined and analyzed.

In the case of Kitchen As-A-Pal, a separate assessment needed to be conducted for adapting the ambient system to Sabina's visual and mobility impairment. The collected information formed the base for a user model of Sabina in relation to the kitchen environment. This model would be used for adapting the interactivity of the physical environment in case Sabina would have used the environment during a longer period. However, in order to collect the amount of information sufficient for creating the activity trajectories specific to Sabina, and comparing these for assessing activity performance, more than two occasions is needed. Consequently, a thorough validation of the use of trajectories is subjected to future work.

7 Related Work

Ubiquitous computing environments are dynamic environments affected by human mobility, ad-hoc availability of computing resources, nearby people, situated objects and the ongoing activity context. Facilitating human interaction with computing systems needs interaction models that take into account both the dynamic and distributed nature of available interfaces in the environment and user models for personalization.

There are existing interaction models for ubiquitous computing that make use of human ability to manipulate physical objects like in tangible user interfaces [53]; that keep track of surrounding environment as in context-aware computing [14]; that use proxemics relations between people and objects [3]; that are activity-based [5]; and general framework for interactions based on inspiration from the physical world like in reality-based interaction [23]. While such approaches are interesting, many of such approaches are device or technology-centric with limited user data making it difficult for personalization and adaptation. Egocentric interaction applied in the As-A-Pal project incorporates individual user's spatial context and associate it to their perception and action abilities useful for personalization in ubiquitous computing environments [46].

Other approaches also take a real world situation as starting point and frame the need for agent-based support [22]. SHARE-IT [17] (Supported human autonomy for recovery and enhancement of cognitive and motor abilities using information technologies) implements a combination of multi-agent system and other techniques to aid the elders based on user scenarios [2]. The agents hold information about all physical devices that exist in the environment to control daily living activities and the conceptual world's information. A difference in our work is that we integrate assessment into the context of use, and aim at using the professionals' instruments for continuing followup performed by software agents in the ubiquitous environment.

As-A-Pal differs from other applications of argumentation-based agreement technologies (e.g., [15,41]) in that it exchanges semantically rich arguments based on a terminology model, which captures uncertainty. This is a necessary property for reasoning about purposeful activities for supporting individuals in their activity performance, using arguments generated by heterogeneous sources in a ubiquitous environment. The handling of uncertainty was also listed as an essential requirement for enabling generic user modeling by Kobsa in [27]. The possibilistic argumentation approach also differentiates As-A-Pal from other ubiquitous user modeling approaches, which typically apply methods based on statistics for handling uncertainty [28].

The actor and activity ontology is partly based on a holistic model of ability, dysfunction and health provided by the World Health Organization, which is familiar to health professionals. This differs our ontology from e.g., the semantic user model presented in [24], which separates the user ontology from the environment ontology, and matches the two for the purpose to identify handicap situations, mainly due to constraints in the physical environment.

To our knowledge, there is no other framework for assistive technology which utilizes a user model, which is based on theories of human occupation in combination with classifications and terminologies from the health domain, this for the purpose to achieve a continuous holistic assessment and service provision to the individual.

8 Discussion

Developing user models for ubiquitous computing environments possesses challenges such as i) the lack of initial user information ("the cold start problem"), ii) heterogeneous sources of user modeling information with limited interoperability, and iii) the need for continuous adaptation of a user model due to changing context [28,10,27], and changing attitudes and needs in the individual.

We have presented a knowledge-based approach to meet these challenges. The "cold start problem" is partially handled by the As-A-Pal architecture through the initial assessments done by therapists, which generate a minimal set of information about a person's health, motives and priorities. This minimal baseline set of information can be used by a system agent for creating a user model suitable for the purpose at hand, for instance, modeling the priorities and goals, which are used for determining interventions as shown in the article. The initial lack of the user information, which the ambient systems need, can also be partially handled by using the derived priority list of activities to be supported, for the purpose to make the initial data collection and refinements for creating a baseline model of typical activity trajectories for a particular user. Ongoing and future work includes the comparison of the trajectories obtained with the different ambient systems, and with professional assessments of motor and process skills. One purpose is to assess the reliability of the

ambient systems' assessments of activity conduction and performance, to be included in the agent-based reasoning about contradictory information.

The reasoning using heterogeneous sources of information is enabled through a common language, the conceptual model for argument and information exchange based on an adaptation of the Argument Interchange Format (AIF) [13] accessed as a service. The arguments and information are instances of classes in the ACKTUS ontologies, which provide the semantics of the observation of, e.g., the human actor performing activities in a home environment. Consequently, interoperability is achieved among the systems which conforms to this common vocabulary.

The third challenge highlighted is the need for continuous adaptation of a user model due to changing context, and attitudes and needs in the individual. While research on adaptive systems typically addresses physical limitations such as visual or mobility impairments, which are often stable over time (e.g., [24]), we address primarily mental conditions in this article. The reason for this is because the major causes of, and obstacles for changing unhealthy living patterns, are mental and neuropsychological conditions. Changing habits and coping strategies requires more mental effort and dedication than continuing with an internalized behavior pattern, which constitutes a part of the habitual system of an individual. To be able to perform and engage in activities both our interests, habits, routines, motivation and self-efficacy are important factors to consider [26]. For this purpose, we explored different personas and scenarios, and conducted a case study, where the potential end user representatives design their support systems, by means of arguments and motives which they believe will have impact on their own behavior. This way we integrate both internal and external motivators, following the Self-Determination Theory (SDT) [43]. Consequently, our approach aims at evoking and reinforcing the individual's own capacity to change behavior, instead of primarily using external sources for imposing change, which the individual may object to.

However, the results regarding effects on motivation obtained in our case study were limited, partly due to the limited time during which the tailored applications were used. In one case, the limitation was due to that the context of use was not fully integrated for the arguments to be timely, and in the other case there were indications that the arguments should be augmented with partial visualizations of the user model for increasing motivation. Since the three participants represented a broad range of priorities and motives, it was also shown that it may be necessary to interpret similar information differently, depending on the user's motives in order to optimize the support.

Based on this, we argue that a careful and evidence-based initial assessment needs to be done, supervised by a therapist in order to interpret the nature of an individual's behavior into a baseline user model, which can subsequently be enhanced, validated and adapted using ubiquitous methods. The As-A-Pal architecture presented in this work integrates the modules for accomplishing this, and will be further evaluated in future user studies.

A limitation of the knowledge-based approach, is that the knowledge integrated into the systems is limited to the knowledge domains, which the domain

experts represent. Moreover, the engineering of such systems is often the bottleneck of the development since it requires time and expertise [28, 44, 48]. We have addressed this problem by introducing ACKTUS, which allows the domain professionals to create knowledge-based web applications and design the content [36, 33]. It should be noted though, that in order to investigate the limitations of the knowledge content, user studies need to be conducted involving a larger number of participants of different age, gender, priorities and needs, which is subjected to future work.

However, a user model is defined in terms of semantics models in our knowledge-based approach, which gives at least the following advantages: i) reasoning mechanisms can be implemented by the architecture in order to support decision making; ii) data-interoperability can be managed between the different sub-systems of the architecture; and iii) given that the semantic models can be extended, our user models are scalable. In other words, the suggested user models can be extended to different user domains.

9 Conclusions

The major contribution of the presented work is an architecture (As-A-Pal) for knowledge-based holistic user modeling in smart environments.

For achieving the holistic perspective, the architecture integrates a generic core ontology based on medical and health terminologies, models of human occupation and a model of egocentric interaction in ubiquitous environments, which was expanded as a part of this work. The core ontology is used for knowledge representation and reasoning, and for sharing information between systems. For interoperability between systems in the architecture, the systems utilize parts of the core ontology as a common vocabulary, and communicate information through a dedicated service.

The As-A-Pal architecture combines the knowledge engineering platform ACKTUS and knowledge-based support systems developed using ACKTUS, with an ambient assisted living environment, which detects pattern of activity and provides tailored support for the execution of activity.

The information used for creating the initial user model is collected by means of the routinely conducted clinical interviews and observations of activity, done by a healthcare professional as part of initial assessments of a client. Based on the information, the system agents create a baseline user model of the user. In this work we investigated in particular how priorities, motives and goals are represented, adjusted and used for determining desired interventions, and how information collected by ambient systems in the environment may contribute to this.

It was exemplified how the model becomes enriched through agent dialogues, in which information from domain knowledge repositories and sources in a ubiquitous environment is combined with subjective information provided by the user. For handling partial information, uncertainty, ambiguities and in-

consistencies, reasoning technologies based on argumentation and answer set programming are used.

A pilot evaluation study was conducted, which demonstrated how the initial assessments formed a base for interventions, though user models capturing prioritized activities and corresponding needs and desire for support. The results showed limitations in how the context of use was integrated, and gave indications on the potentials in using visualizations of the information obtained by the ambient systems in combination with own-created arguments for increasing motivation in the user.

Future work includes an evaluation study conducted over a longer period of time, for the purpose to explore how the support needs to be adjusted to changing motives, goals and priorities in an individual. Further implementation work is being done, including improving the timing of feedback and support by integrating more contextual information, and improving the agent dialogues, partly by integrating dialogues for persuasion and negotiation. Moreover, different approaches to presenting relevant parts of the user model in different situations and for different purposes will be investigated.

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